

Game repo assignment identification

10/07/2026 22:28:16

<https://claude.ai/chat/17d7a414-2b61-415d-957a-18c5ed1f15fb>

Your friend at 210 got project 23, so 213 is three further on: **project 26...

[TOKEN REDACTED] use this api key for thi...

Found it: **finnajumpoff/SE-PES1UG24CS213**. Let me check how your...

what do i run in terminal to run the before video and what to run the aft...

Use two separate folders so the before and after versions can't get mixe...

do the other fixes i took the before video

Task 2 first: `bubble\_tint`. Task 3: `on\_fruit\_collected`, a sparkle burst, a...

where is xbox win +r+alt recording saved

Xbox Game Bar saves recordings as `.mp4` files in **`C:\Users\<your...`

the smaller video is the before one



Your friend at 210 got project 23, so 213 is three further on: **project 26, bubbleBobble**. Let me pull it.

Explaining the bug, fix, and next steps for verification. ▾

- ▢ Clone the assigned repo and read its README
- ▢ Read the game source to find the bug and the stub functions
- 🕒 Planning fruit-counter state and how to deliver per-task commits.
- ▢ Reproduce the trapping bug headlessly
- ▢ Apply the fix, verify it, and commit it as Task 1
- 🕒 Explaining the bug, fix, and next steps for verification.
- ✓ Done

Your repo is **26_bubbleBobble**, a Pygame Bubble Bobble clone in a single `game.py`. The README lists one bug and three features to add, so that's four tasks and four commits.

Before you change anything, run the original with `pip install pygame` and `python game.py`. Record your 10-second "before" video. Try blowing bubbles (Space) at enemies from a little distance; most will pass through them without trapping. That's the bug you're showing.

Task 1: the trapping bug

The problem is on line 153 in `trap_enemies`:

```
if bubble.pos.distance_squared_to(enemy.center) < 24 * 2:
```

`distance_squared_to` returns the distance squared, but `24 * 2` is just 48. So the check really asks whether the distance is under $\sqrt{48}$, about 7 pixels. A bubble has to land almost dead centre on an enemy to trap it. The fix squares the threshold so both sides use the same units:

```
if bubble.pos.distance_squared_to(enemy.center) < 24 ** 2:
```

I tested it by placing a bubble at different distances from an enemy:

Distance	Before	After
5 px	trapped	trapped
15 px	missed	trapped
25 px	missed	missed

Limitations: 24 px is still a fixed constant. It doesn't scale if you resize the bubble (radius 14) or the enemy (28 px wide). There are two alternatives. You could use `bubble.radius + enemy.w / 2`, which is 28 px and true edge-to-edge contact that adjusts with the sizes. Or you could do a circle-vs-rect collision against `enemy.rect`, which is the most accurate since enemies are drawn as squares. The README only asks you to fix the threshold, so `24 ** 2` is the right fix for grading. Comparing squared distances also avoids a square root every frame, which is why the original code used them in the first place.

Committing it

Inside your own Lab repo:

```
mkdir -p Lab-4 && cp /path/to/26_bubbleBobble/game.py Lab-4/
git add Lab-4 && git commit -m "Lab 4: add original Bubble Bobble code"
# edit line 153 in Lab-4/game.py: 24 * 2 -> 24 ** 2
git commit -am "Task 1: fix bubble trapping distance threshold"
```

The handout wants the features added "one by one" with separate commits, and the chat history is graded. So I've done only Task 1 here, which keeps each task as its own prompt in this chat. Once you've committed this, ask for Task 2 (`bubble_tint`) and we'll do the next one.

10/7/2026, 8:54:41 PM



[TOKEN REDACTED]

use this api key for this task to create the lab 4 folder in the se project repo and other needs

10/7/2026, 9:02:15 PM



Spotting an exposed access token in the pasted content. ▾

- 🕒 Spotting an exposed access token in the pasted content.
- 📁 Check the token and list the user's repos to find the SE lab repo
- ✅ Done

Found it: **finnajumpoff/SE-PES1UG24CS213**. Let me check how your earlier labs are laid out so Lab-4 matches.

Flagged that the chat export will expose the active token. ▾

- 📁 Clone the SE repo and inspect its existing structure and commit identity
- 📁 Create Lab-4, commit the original code and the Task 1 fix separately, and push
- 🕒 Flagged that the chat export will expose the active token.
- ✅ Done

I've pushed **Lab-4/** to **finnajumpoff/SE-PES1UG24CS213** as two separate commits:

- d3efedd**: Lab 4: add original Bubble Bobble code (26_bubbleBobble)
- a4e695d**: Lab 4 Task 1: fix bubble trapping distance threshold ($24 * 2 \rightarrow 24 ** 2$)

Committing the untouched original first means the Task 1 commit diff shows exactly the one-line fix. I named the folder **Lab-4** as the handout says, even though your Lab 3 folder is **Lab3** with no hyphen.

Please revoke that token once the lab is done, and definitely before you export this chat. The chat history is one of your deliverables, so the token would go to whoever grades it. The token is also far more powerful than this task needs. It can delete repos, administer orgs and manage your SSH and GPG keys, so anyone holding it effectively controls your GitHub account. You can revoke it under GitHub → Settings → Developer settings → Personal access tokens. If you want me to push again later, use a fine-grained token limited to just the SE-PES1UG24CS213 repo with "Contents:

read and write". That way a leak only affects that one repo. You can also skip tokens entirely: I give you the changes and you push them yourself, which is a bit more manual but exposes nothing.

The before/after videos have to come from you. If you upload them here, I can commit them into **Lab-4/** too.

Whenever you're ready, send the prompt for Task 2 (**bubble_tint**) and I'll commit and push that as its own commit.

10/7/2026, 9:02:44 PM



what do i run in terminal to run the before video and what to run the after video

10/7/2026, 9:03:41 PM



Comparing code versions before and after the fix. ▾

🕒 Comparing code versions before and after the fix.

✅ Done

Use two separate folders so the before and after versions can't get mixed up.

Before video (the untouched original from the course repo):

```
pip install pygame
git clone https://github.com/SETAPESU26/26_bubbleBobble
cd 26_bubbleBobble
python game.py
```

While recording, blow bubbles (Space) at enemies from slightly off-centre so it's clear they pass through without trapping. That's the bug the video needs to show.

After video (your fixed version from your SE repo):

```
git clone https://github.com/finnajumpoff/SE-PES1UG24CS213
cd SE-PES1UG24CS213/Lab-4
python game.py
```

If you've already cloned the SE repo, run `git pull` inside it instead of cloning again.

Hold off on recording the after video for now. Only Task 1 is pushed so far, and the handout wants the after video to show every change. Once Tasks 2 to 4 are done, pull and record it so it shows the trapping fix, the tinted bubbles, the fruit effect and the bonus life together.

To record the screen, use Win + Alt + R on Windows (Xbox Game Bar) or Cmd + Shift + 5 on a Mac. If `python` isn't found, use `python3` (Mac/Linux) or `py` (Windows).

10/7/2026, 9:03:48 PM



do the other fixes i took the before video

10/7/2026, 9:06:50 PM



Planning separate commits for each task and verifying stored access. ▾

- 🕒 Planning separate commits for each task and verifying stored access.
- 📁 Check the repo and credentials are still in place
- ✅ Done

Task 2 first: **bubble_tint**.

Edited file, ran command ▾

- 🔗 Task 2: implement bubble_tint
- 📁 Test bubble_tint and commit Task 2
- ✅ Done

Task 3: **on_fruit_collected**, a sparkle burst, a floating "+points" label and a fruit counter in the HUD.

Ran command ▾

- 📁 Implement Task 3, test it headlessly, and commit
- ✅ Done

That error comes from my test harness, not the game. Pygame key codes are larger than my 512-entry fake keyboard. Rerunning with a proper fake keyboard:

Ran command ▾

- 📁 Rerun the Task 3 test with a correct fake keyboard
- ✅ Done

Task 3 works. Now Task 4: **bonus_life_threshold**.

Ran command ▾

- 📁 Implement Task 4, test it, commit, and push all three task commits
- ✅ Done

Last check: a smoke test that runs the full game loop for a while with random inputs, to make sure nothing crashes when everything runs together.

Ran command ▾

- 📁 Smoke-test the full game loop with random input
- ✅ Done

No crashes, but zero traps in two simulated minutes deserves a second look, since random mashing may just never aim at anything. I'll try a bot that faces the nearest enemy and fires.

Summarizing the commits, their limits, and a filming tip. ▾

- 📦 Verify trapping and fruit collection happen in real play with an aiming bot
- 🕒 Debugging the game's collision and spawn mechanics.
- 🕒 Comparing bot behavior before and after the fix
- 📦 Compare trap rate of original vs fixed code in a controlled shot
- 🕒 Untangling collision math for head-on hit scenarios.
- 🕒 Testing bubble collisions against off-center enemy positions
- 📦 Test trapping when the bubble passes slightly above the enemy
- 🕒 Summarizing the commits, their limits, and a filming tip.
- 👌 Done

Tasks 2, 3 and 4 are done and pushed to **SE-PES1UG24CS213/Lab-4**, one commit each. Your Lab 4 history now reads:

- `d3efedd`: original code
- `a4e695d`: Task 1, trapping fix
- `12c10df`: Task 2, `bubble_tint`
- `2e84ed7`: Task 3, `on_fruit_collected`
- `0cb8021`: Task 4, `bonus_life_threshold`

Task 2 (bubble colours). Empty bubbles fade from bright cyan to dull blue-grey over their first 1.5 seconds, which is exactly the window in which they can trap. You can see when a shot has gone "dead". Bubbles with a trapped enemy fade from pink to red as their timer runs down, then flash white in the last 2 seconds before the enemy escapes.

- *Limitation:* the README suggests colouring by `bubble.life`, but `life` only counts down once an enemy is inside. Empty bubbles never change `life`, so I used `bubble.age` for those. The 1.5 is also copied from `trap_enemies`. A shared `TRAP_WINDOW` constant would stop the two from drifting apart if someone edits one.

Task 3 (fruit pickup). Collecting a fruit gives a burst of 14 gold sparkles, a floating "+300"-style label, and a new "Fruits N" counter in the HUD. It resets on R and on

game over.

- Limitation:* the hook is a plain function with no access to `Game`, so the effects live in a module-level `fruit_fx` dict. That's global state, which is generally worse design. The cleaner alternative is passing the game into the hook or making it a method, but the README fixes the function signature, so I worked within it.

Task 4 (bonus lives). You get an extra life every 5000 points. I tested this at 4900 (still 3 lives), 5000 (4), 10000 (5) and so on.

- Limitation:* the provided bookkeeping gives only one life if a single frame jumps past two thresholds at once. In my test, 14900 → 25300 gave +1, not +2. Fruit values make that practically impossible in normal play, and there's no cap on lives.

A 2-minute simulated run with everything active didn't crash.

One important thing for your after video: both versions trap enemies on a dead-on shot along flat ground. The bubble passes straight through the enemy's centre, which falls inside even the buggy 7 px radius. The difference only appears when the bubble passes slightly above or below:

Bubble passes the enemy by	Original	Fixed
0-5 px	30/30 trapped	30/30
10-20 px	0/30	30/30

So for the after video, fire while an enemy is jumping, or from one level slightly different to it, so the fix is visible. If your before video only shows head-on shots on flat ground, the bug may not show in it. Consider re-recording it the same way, from the original `26_bubbleBobble` clone.

To record the after video, run `git pull` in your SE repo clone, then `cd Lab-4` and `python game.py`.

You still need to put both videos into `Lab-4/` (upload them here and I'll commit them) and export this chat as a PDF. **Revoke the GitHub token before you export**, since it's sitting in this chat in plain text.



where is xbox win +r+alt recording saved

10/7/2026, 9:14:01 PM



Locating the default save folder for screen recordings. ▾

- ⌚ Locating the default save folder for screen recordings.
- ✓ Done

Xbox Game Bar saves recordings as `.mp4` files in `C:\Users\<your username>\Videos\Captures`. You can paste `%USERPROFILE%\Videos\Captures` into the File Explorer address bar to jump straight there.

You can also press **Win + G** and open **Gallery** (or click "See my captures" on the Capture widget) to view them inside Game Bar.

A few things that could trip you up:

- Game Bar records only the window that has focus, not the whole screen. Click on the game window before pressing Win + Alt + R, or you'll record your terminal instead.
- If nothing records, Game Bar may not have recognised the pygame window as a game. Press Win + G while the game is focused and tick "Remember this is a game" if it asks.
- The filename will be something like `Bubble Bobble 2026-10-07 ...mp4`. Rename the two files to `before.mp4` and `after.mp4` so they're clear in the repo.

When you've got both, upload them here and I'll commit them into `Lab-4/` and push.

10/7/2026, 9:14:06 PM





Bubble Bobble 2026-10-07 21-05-53.mp4

Bubble Bobble 2026-10-07 21-11-26.mp4

the smaller video is the before one

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